Sahil Singh

sahilsingh107433@gmail.com — +91 73183 84207 — github.com/sadsanta-07 — linkedin.com/in/sahilsingh107433

Summary

Software Engineer with a strong foundation in C++, Python, and full-stack web technologies (React, Node.js). Experienced in architecting distributed systems, concurrent backends, and AI integrations. Built production-ready applications featuring robust SQL databases, real-time web pipelines, and thread-safe APIs. Passionate about solving ambiguous abstract problems, optimizing algorithmic complexity, and building scalable enterprise tools.

Projects

OfflineUPIMesh – Distributed Offline Payment Engine *C++20, Crow, ASIO, OpenSSL, SQLite3, CMake*

- Architected a high-performance C++ microservice for mesh-routed offline payments, demonstrating deferred ledger settlement and secure distributed systems concepts across untrusted intermediary nodes.
- Engineered a highly concurrent ingestion pipeline utilizing `std::mutex` locks and `unordered_map` hashing to instantly short-circuit duplicate packet storms (O(1) lookups), efficiently preventing double-spending.
- Implemented TLS-style hybrid cryptography (AES-256-GCM wrapped in RSA-2048) to guarantee payload authenticity, and ensured ACID-compliant financial debits/credits via strict SQLite exclusive transactions.

NoJudgement – AI Speaking Practice Platform *Next.js (React), WebRTC, PostgreSQL, Python (LLaMA)*

- Developed a scalable web application integrating WebRTC and Socket.io for live peer matchmaking, successfully handling concurrent user sessions and fixing race-condition anomalies on singleton sockets.
- Designed relational database schemas in PostgreSQL (Prisma ORM) to support automated reporting pipelines, streak tracking, and gamification leaderboards at scale.
- Built chunk-based real-time STT pipelines integrating Whisper and LLaMA for instant, AI-driven grammar feedback, dynamically rendering technical solutions into an intuitive frontend UI.

NanoGPT – Decoder-Only Transformer From Scratch *Python, PyTorch*

- Implemented a GPT-style transformer entirely from scratch—including causal self-attention, masking, positional embeddings, and residual connections—without relying on high-level HuggingFace abstractions.
- Trained a 6-layer, 384-dim model via autoregressive generation, developing a deep intuition for advanced data structures, matrix operations, and complex algorithmic architectures under the hood.

Experience

Software Developer Intern May 2026 – Present
IFFCO (Indian Farmers Fertiliser Cooperative) *Phulpur, Uttar Pradesh*

- Contributing to enterprise-scale internal software systems, leveraging modern full-stack workflows and relational database architectures to support critical business operations.
- Assisting in backend API development, automated testing workflows, and modular architecture design to improve system monitoring and overall operational efficiency.
- Collaborating cross-functionally using agile methodologies to effectively communicate technical challenges and implement optimized solutions in a fast-paced environment.

Web Developer Feb 2026 – Present
Google Developer Groups KIIT Chapter *Bhubaneswar, Odisha*

- Architected responsive, event-driven web applications using React, HTML/CSS, and Node.js for a 500+ member tech community, coordinating closely with peers to consistently ship features on deadline.
- Integrated RESTful APIs and PostgreSQL/Supabase authentication, optimizing upload and data delivery workflows.

Technical Team Member – Web Developer May 2025 – Present
K-1000 *Bhubaneswar, Odisha*

- Maintained scalable backend architecture supporting full CRUD operations; refactored legacy API routes into modular server actions to decrease latency and enhance code maintainability.

Education

KIIT University **Class of 2028** (Aug 2024 – Present)
B.Tech – Computer Science (AI & ML) *Bhubaneswar, Odisha*

Technical Skills

Languages	C/C++, Python, Java, JavaScript, TypeScript, SQL
Web / Backend	React, Next.js, Node.js, Express.js, HTML/CSS, WebRTC, REST APIs, Socket.io
Databases	PostgreSQL, SQLite, MySQL, MongoDB, Supabase, Prisma ORM
CS Fundamentals	Object-Oriented Design, Data Structures & Algorithms, Distributed Systems, Concurrency, ACID
Tools / ML	Git, GitHub, Linux, CMake, Docker, PyTorch, Postman, Vercel